1. What is IoT?

Answer: IoT stands for Internet of Things. It connects physical devices to the internet so they can collect, exchange, and process data.

2. What are the main components of IoT?

Answer: The main components are:

Sensors

Processing unit

Communication/network

Cloud or server

Actuators

3. What is a Sensor?

Answer: A sensor is a device that detects physical conditions such as temperature, light, humidity, or motion and converts them into data.

4. What is an Actuator?

Answer: An actuator is a device that performs an action based on a command. For example, a motor can turn on when an IoT system sends a signal.

5. What is an IoT device?

Answer: An IoT device is a physical device that can collect data, communicate over a network, and sometimes perform actions automatically.

6. What is the role of the Internet in IoT?

Answer: The Internet allows IoT devices to communicate with servers, cloud platforms, and other devices and exchange data.

7. What is MQTT?

Answer: MQTT is a lightweight communication protocol commonly used in IoT. It allows devices to exchange messages efficiently, especially when network resources are limited.

8. What is Cloud Computing in IoT?

Answer: Cloud computing provides remote storage, processing, and analysis of data collected from IoT devices.

9. What are the applications of IoT?

Answer: IoT is used in:

Smart homes

Smart agriculture

Healthcare

Smart cities

Industrial automation

Environmental monitoring

10. What are the challenges of IoT?

Answer: Major challenges include:

Security and privacy

Large amounts of data

Network connectivity

Power consumption

Device management

Compatibility between different devices